Comparative Analysis of Radio Network Planning Reports

Kagal vs. Ralegaon vs. Pombhurna vs. Sakri

4G LTE Cellular Network Design — 3GPP TR 38.901 V18.0.0

Auto-Generated Comparison Report

April 5, 2026

Abstract

This document presents a section-by-section comparison of four independently authored radio network planning reports for the same assignment. All four reports design a 4G LTE network for a different town in Maharashtra using 3GPP TR 38.901 V18.0.0 and then simulate small-scale fading via the Clustered Delay Line (CDL) channel model. The towns are: **Kagal** (Kolhapur district) by Aditya Kinjawadekar, **Ralegaon** (Yavatmal district) by Atharva S Chaudhari, **Pombhurna** (Chandrapur district) by Prasad Patil (BT23ECE096), and **Sakri** (Dhule district) by Tanmay Lakhambare (BT23ECE121). Each section highlights agreements, disagreements, and unique features across the four reports.

Contents

1 Report Metadata and Formatting **2**

2 Site Analysis and Location Overview **3**

3 System Parameters and Link Budget **3**
3.1 Link Budget Calculation — Critical Difference . . . 4

4 Path Loss Models **5**

5 Maximum Cell Radius ($R_{\max}$) and D/R Design **5**

6 Network Design and Iterations **6**

7 Microcell and Umbrella Cell Justification **7**

8 Part 2: Small-Scale Fading / Channel Simulation **8**

9 Python Code Comparison **9**

10 Comprehensive Numerical Comparison **10**

11 Summary of Differences and Assessment **12**
11.1 Strengths by Report . . . 12
11.2 Identical Content Alert . . . 13
11.3 Breakpoint Distance Error in Sakri Report . . . 13
11.4 Overall Assessment . . . 13

1 Report Metadata and Formatting

Table 1: Report Metadata Comparison

Attribute	**Kagal (Kinjawadekar)**	**Ralegaon (Chaudhari)**	**Pombhurna (Patil)**	**Sakri (Lakhambare)**
Town / Area	Kagal, Kolhapur	Ralegaon, Yavatmal	Pombhurna, Chandrapur	Sakri, Dhule
Population (est.)	≈49,000 (2026)	≈18,500 (2026 est.)	≈20–25,000 (2011)	≈32,400 (2026 est. from 21,764 in 2011)
Town Area	≈8–10 km^2	18.2 km^2	≈12 km^2	≈6.75 km^2
Elevation	553 m	248 m	230–250 m	Not stated
Document Format	LaTeX 2-column, 10pt	PDF, single-column	PDF, single-column	LaTeX 2-column
Total Pages (approx.)	∼9 (body) + appendix	17 pages	16 pages	13 pages
Appendix Code?	Yes (2 Python listings)	No separate appendix	Yes (code in body)	No separate appendix
Figures Included?	6 site photos + 3 sim plots	2 maps	1 site analysis map	14 figures (satellite, maps, site photos, network diagrams)

Key Observations:

- Kagal uses professional LaTeX two-column formatting; Sakri also uses LaTeX two-column; the other two are single-column PDF reports.
- Kagal includes street-level site photography; Ralegaon includes KML maps; Pombhurna includes a schematic; Sakri includes extensive figures (satellite views, market, bus stand, river, schools, hospital, and all network iteration diagrams).
- Sakri is the only report that includes figures for every step of the network design iteration process.
- All four cover the same two-part assignment: Part 1 (network planning) and Part 2 (channel matrix / CDL simulation).

2 Site Analysis and Location Overview

Table 2: Site Analysis Comparison

Feature	**Kagal**	**Ralegaon**	**Pombhurna**	**Sakri**
Coordinates	16°35′N, 74°19′E	20°25′N, 78°31′E	Near 20°N, 79°E	Sakri, Dhule (approx. 20°56′N, 74°18′E)
Terrain Type	Semi-urban, 2–3 storey	Hot tropical, cotton land	Flat, Deccan Plateau	Residential + agricultural fields
Key Highway	NH-48 (6-lane, western edge)	State Highway (diagonal)	NH-930 (through core)	NH-53 Surat–Kolkata (near town)
Key Water Body	Dudhganga River	None prominent	Penganga River (3–5 km SW)	Panzara River (southern boundary)
Commercial Area	Market / town centre	Main market & mills	Main bazaar along NH-930	Markets, hospitals, bus stand along Sakri Road
Building Heights	6–10 m (2–3 storeys)	6–9 m (1–3 storeys)	2–3 floors (centre)	Not explicitly stated
Coverage Zones	6 identified	4 identified	5 identified	3 primary: macrocell, microcell (market/bus), umbrella (highway)
Visual Evidence	4 street-level photos	KML maps	Satellite + schematic	Satellite + 7 site photos (market, bus, river, schools, hospitals)

Differences:

- **Kagal** is the most densely populated (≈49k) and has the smallest area (≈8–10 km^2).
- **Ralegaon** has the largest area (18.2 km^2) but smallest population (≈18.5k).
- **Sakri** has the smallest coverage area (≈6.75 km^2, defined by Panzara River and NH-53 boundary) and a moderate population (≈32.4k). It explicitly measures the perimeter (10.11 km), max horizontal extent (4097 m), and highway length within coverage (5626 m).
- Kagal and Sakri both provide street-level photographs; Ralegaon uses KML waypoints; Pombhurna uses schematic diagrams.

3 System Parameters and Link Budget

This is the most critical section for comparison, as the link budget directly determines the maximum cell radius.

Table 3: System Parameters Comparison

Parameter	**Kagal**	**Ralegaon**	**Pombhurna**	**Sakri**
Carrier Frequency f_c	1800 MHz	1800 MHz	1800 MHz	1800 MHz
Transmit Power P_t	46 dBm	46 dBm	46 dBm	46 dBm
BS Antenna Gain G_{ant}	18 dBi	18 dBi	17 dBi	Not in link budget
Cable / Feeder Loss	2 dB	2 dB	3 dB	Not in link budget
Rx Sensitivity $P_{r,\min}$	−104 dBm	−104 dBm	−94 dBm (computed)	−104 dBm
UE Antenna Gain	—	—	0 dBi	—
Body Loss	—	—	3 dB	—
Noise Figure	—	—	7 dB	—
Indoor Penetration Loss	—	—	10 dB	—
Shadow Fading Margin L_{sf}	8 dB	8 dB	8 dB	10 dB
BS Height h_{BS}	25 m	25 m	25 m	25 m
UT Height h_{UT}	1.5 m	1.5 m	1.5 m	1.5 m
Bandwidth	—	—	10 MHz	—
Min SNR	—	—	3 dB (QPSK 1/3)	—

3.1 Link Budget Calculation — Critical Difference

Table 4: Link Budget Comparison — $L_{\max}$ Derivation

Step	**Kagal**	**Ralegaon**	**Pombhurna**	**Sakri**
Simple $L_{\max}$	$46 - (-104) = 150$ dB	$46 - (-104) = 150$ dB	Not computed separately	$46 - (-104) = 150$ dB
Extended formula	$P_t + G_{\text{ant}} - L_{\text{cable}} - P_{r,\min} - L_{\text{sf}}$	Same	EIRP $+ G_{\text{UE}} - L_{\text{body}} - M_{\text{sf}} - L_{\text{pen}} - P_{r,\min}$	$L_{\max} = P_t - P_{r,\min} - L_{\text{sf}}$
Substitution	$46 + 18 - 2 + 104 - 8$	Same	$60 + 0 - 3 - 8 - 10 - (-94)$	$150 - 10$
Result $L_{\max}$	**158 dB**	**158 dB**	**133 dB**	**140 dB**

Critical Differences:

- **Four different $L_{\max}$ computation approaches** are now visible across the four reports:
  1. **Kagal & Ralegaon** ($L_{\max} = 158$ dB): Use antenna gain and cable loss in the extended budget.
  2. **Pombhurna** ($L_{\max} = 133$ dB): Uses thermal noise floor with NF, body loss, and penetration loss.
  3. **Sakri** ($L_{\max} = 140$ dB): Uses the simplest approach — just $P_t - P_{r,\min} - L_{\text{sf}} = 150 - 10 = 140$ dB. Does **not** include antenna gain or cable loss in the link budget.
- Sakri uses a fading margin of **10 dB** (vs. 8 dB for the other three), citing it as a practical worst-case margin exceeding the 6 dB σ_{SF} from 3GPP.
- The spread is now 25 dB between the highest (158 dB) and lowest (133 dB) $L_{\max}$ values.

4 Path Loss Models

Table 5: Path Loss Model Comparison

Aspect	Kagal	Ralegaon	Pombhurna	Sakri
Primary Model	UMa NLOS	UMa NLOS	UMa NLOS	UMa NLOS
3GPP Reference	TR 38.901 Table 7.4.1-1	Same	Same	Same
UMa NLOS formula	$13.54 + 39.08\log_{10}(d_{3D}) + 20\log_{10}(f_c) - 0.6(h_{UT} - 1.5)$	Same	Same	Same
σ_{SF} (UMa NLOS)	6.0 dB	Not stated	Not stated	6 dB (mentioned, but uses 10 dB margin)
UMi NLOS (microcell)	Yes, presented	Not presented	Not presented	Not presented
RMa NLOS (umbrella)	Referenced	Used for 13 umbrella cells	Not presented	Not presented
Breakpoint d'_{BP}	452.4 m	452.4 m	Not computed	900 m
UMa LOS formula	Fully stated (2 regimes)	Fully stated	Not stated	Fully stated (2 regimes)
NLOS justification	Dense residential NLOS photos	Conservative worst-case	Conservative worst-case	Semi-urban environment

Observations:

- All four use the same UMa NLOS formula from 3GPP TR 38.901.
- **Breakpoint distance discrepancy:** Kagal and Ralegaon compute $d'_{BP} = 452.4$ m (using effective heights $h'_{BS} = 24$ m, $h'_{UT} = 0.5$ m as per 3GPP TR 38.901). **Sakri computes $d_{BP} = 900$ m** using the raw heights ($h_{BS} = 25$ m, $h_{UT} = 1.5$ m) without the 3GPP effective height correction. This is a **methodological error** — 3GPP TR 38.901 defines $h'_{BS} = h_{BS} - h_E$ and $h'_{UT} = h_{UT} - h_E$ with $h_E = 1$ m for UMa, giving $d'_{BP} = 452.4$ m, not 900 m.
- **Kagal** is the only report that explicitly presents the UMi NLOS model for microcell planning.
- **Sakri** uniquely takes $\text{PL} = \max(\text{PL}_{\text{LOS}}, \text{PL}_{\text{NLOS}})$ for final path loss evaluation, which is correct per the standard.

5 Maximum Cell Radius ($R_{\max}$) and D/R Design

Table 6: $R_{\max}$ and Network Sizing Comparison

Parameter	Kagal	Ralegaon	Pombhurna	Sakri
$L_{\max}$ used	158 dB	158 dB	133 dB	140 dB
$R_{\max}$ (UMa NLOS)	3680 m	3680 m	843 m	Not computed explicitly
$R_{\max}$ (UMi NLOS)	4868 m	—	—	—
$R_{\max}$ (RMa NLOS)	6258 m	6258 m	—	—
Reuse Factor N	1 (OFDMA + ICIC)	7	7	7
D/R ratio	$\sqrt{3} \approx 1.732$	$\sqrt{21} \approx 4.583$	$\sqrt{21} \approx 4.583$	$\sqrt{21} \approx 4.583$
$R_{\max}$ derivation	Analytic closed-form	Analytic closed-form	Binary search (code)	Design-then-verify approach
Verification shown?	Yes (PL at 3680 m = 158 dB)	Yes	Yes (PL vs. distance table)	Yes (PL at each radius checked)

Critical Differences:

- Kagal uses $N = 1$ (unit frequency reuse for LTE OFDMA with ICIC), giving $D/R = \sqrt{3}$. The other three use $N = 7$, giving $D/R = \sqrt{21} \approx 4.583$.
- **Sakri does not compute $R_{\max}$ explicitly.** Instead, it selects cell radii from practical ranges (800 m for macro, 300 m for micro, 1500 m for umbrella), then verifies whether the path loss at those radii is below $L_{\max} = 140$ dB. This is an **inverse approach** compared to the other three reports that compute $R_{\max}$ first and then design cells within that limit.
- Pombhurna's $R_{\max}$ is 4.4× smaller than Kagal/Ralegaon due to the conservative link budget.

6 Network Design and Iterations

Table 7: Final Network Design Comparison

Design Aspect	**Kagal**	**Ralegaon**	**Pombhurna**	**Sakri**
Number of Iterations	3 (3→5→7 sites)	2	2	2 (initial → umbrella redesign)
Iteration 1	3 sites (gaps found)	8 linear sites (dead zones)	7 cells, varied radii	Full plan: macro + micro + umbrella (R = 1500 m fails PL)
Final Design:				
Total Macrocell Sites	7	4 normal + 13 umbrella = 17	8 normal	3
Macrocell Radius	800 m (uniform)	750 m (normal)	800 m (uniform)	800 m
Umbrella Cells	0 (hex overlap covers highway)	13 ($R = 1000$ m)	1 ($R = 1000$ m)	2 ($R = 1200$ m, redesigned from 1500 m)
Microcells	2 ($R = 200$–300 m)	0	1 ($R = 200$ m)	Zone microcells ($R = 300$ m)
Cell Radius Approach	Uniform R for all macros	Two tiers: 750 m + 1000 m	Uniform R for macros	Fixed R per tier
PL at cell edge	132.1 dB	131.0 dB / 135.9 dB	132.1 dB	132 dB (macro) / 138 dB (umbrella)
Margin vs. $L_{\max}$	25.9 dB	27.0 dB / 22.1 dB	0.89 dB	8 dB (macro) / 2 dB (umbrella)
Total coverage area	11.6 km^2	≈39.6 km^2	≈13.3 km^2	≈6.75 km^2
GPS Coordinates?	No	Yes (17 lat/lon from KML)	No	No

Key Differences:

- **Ralegaon** has the most aggressive design: 17 total sites for 18.2 km^2.
- **Kagal** has a clean 7-cell hex ring with no umbrella cells; the highway is covered by macrocell overlap.
- **Sakri** has a practical iterative design process: initial umbrella cells at $R = 1500$ m **fail** the path loss constraint (144 > 140 dB), so they are redesigned to $R = 1200$ m (138 < 140 dB). This is well-documented with supporting figures for every iteration.
- **Pombhurna** has almost no margin (0.89 dB); **Sakri** has only 2 dB margin on its umbrella cells.
- Sakri uniquely provides exact path loss calculations for both LOS and NLOS at each cell radius, taking $\mathrm{PL} = \max(\mathrm{PL}_{\mathrm{LOS}}, \mathrm{PL}_{\mathrm{NLOS}})$.

7 Microcell and Umbrella Cell Justification

Table 8: Special Cell Layer Comparison

Cell Type	Kagal	Ralegaon	Pombhurna	Sakri
Microcells				
Count	2	0	1	Zone microcells (count unspecified)
h_{BS}	10 m	—	10 m	Not stated
Radius	200–300 m	—	200 m	300 m
Model	UMi NLOS	—	UMi NLOS	Not explicitly stated
Location	Market + Bus stand	—	Market / NH-930	Market, bus stand, Sakri Road corridor
PL verification	Yes (115.3 dB at 300 m)	—	Mentioned, not computed	Not computed for microcells
Umbrella Cells				
Count	0	13	1	2
h_{BS}	—	35 m	35 m	Not stated (implied standard)
Radius	—	1000 m	1000 m	1200 m (redesigned from 1500 m)
Model	—	RMa NLOS	RMa NLOS	UMa (same model; no separate RMa)
Purpose	N/A (hex overlap)	Highway + agricultural	Highway overlay	NH-53 highway coverage
Justification depth	N/A	2 full subsections	1 paragraph	Full iterative redesign with PL verification

Observations:

- Ralegaon's use of 13 umbrella cells remains architecturally unusual.
- **Sakri's umbrella cell redesign** is the most methodologically rigorous: the initial $R = 1500$ m is shown to fail (PL $= 144 > 140$ dB), prompting a redesign to $R = 1200$ m (PL $= 138 < 140$ dB). Both calculations are shown step-by-step.
- Kagal eliminates the need for umbrella cells entirely through a 7-cell hex ring design.
- **Sakri uses the same UMa model for umbrella cells** instead of shifting to RMa NLOS as the other reports do. This is noteworthy because UMa is intended for $h_{BS} = 25$ m in urban macro scenarios, while umbrella cells typically use taller towers (35 m) with the RMa model.

8 Part 2: Small-Scale Fading / Channel Simulation

Table 9: Part 2 — CDL Channel Simulation Comparison

Aspect	Kagal	Ralegaon	Pombhurna	Sakri
3GPP Reference	Section 7.5, TR 38.901	Same	Same	Same
Steps Implemented	All 12 steps	Steps 1–6 (num) + 7–12 (disc)	Steps 1–6 (num) + 7–12 (disc)	Steps 1–6 (numerical + code results)
MIMO Config				
BS Antennas	8 (4-col cross-pol)	4 (ULA)	4 (ULA)	1 (single BS, simplified)
UE Antennas	2	2	2	1 (single UT)
H matrix size	2×8	2×4	2×4	Not applicable (link-level only)
Simulation Distance	583 m	500 m	500 m	632 m
Number of Clusters	20	20	3	19 (UMa NLOS per TR 38.901)
Rays per Cluster	20	20	Not specified	Not specified
r_τ (delay scaling)	2.3	2.3 (code) / 2.1 (text)	2.1	Not stated
C_ϕ	1.289	1.289	Not stated	Not stated
C_θ	1.178	1.178	Not stated	Not stated
LSP Parameters:				
$\mu_{\log_{10}(\mathrm{DS})}$	-6.955	-6.955	-6.332	-6.89
ASA μ	1.81	1.81	Not stated	1.76
ASD μ	1.59	1.59	Not stated	1.62
ZSA μ	0.88	0.88	Not stated	0.70
ZSD μ	0.50	0.50	Not stated	0.47
Polarisation	Cross-polarised (Eq. 7.5-22)	VV only	VV only	Simplified omnidirectional
Doppler included?	Yes ($v = 3$ km/h)	Yes (disc: $v = 50$ km/h)	Yes (discussion)	Mentioned ($v = 3$ km/h)
Output Results				
SVD analysis	Not shown	$\sigma_1 = 1.94$, CN $= 2.87$	Not shown	Not shown
Capacity estimate	Not shown	1.82 bps/Hz	Not shown	Not shown
Plots	PDP, APS, $\lvert H \rvert$ heatmap	None	None	None
Code results shown?	Yes (full output)	In body	In body	Yes (LSP values, cluster delays, cluster powers from code)

Key Differences:

- Kagal implements **all 12 steps** with a full 2×8 MIMO system, cross-polarised antennas, Doppler, and produces PDP/APS/heatmap plots. Most complete implementation.
- **Ralegaon and Pombhurna** implement Steps 1–6 numerically and discuss Steps 7–12 qualitatively.
- **Sakri** implements Steps 1–6 with **Python code output** (cluster delays, cluster powers, LSPs) but uses a simplified **single-antenna link-level model** (1 BS, 1 UT). It does not construct a MIMO channel matrix.
- Sakri uses $N = 19$ clusters (correct for UMa NLOS per 3GPP), matching the standard better than Pombhurna's $N = 3$.

- **Sakri's LSP values differ** from Kagal/Ralegaon: $\mu_{\mathrm{DS}} = -6.89$ vs. -6.955; ASA $\mu = 1.76$ vs. 1.81; ASD $\mu = 1.62$ vs. 1.59; ZSA $\mu = 0.70$ vs. 0.88; ZSD $\mu = 0.47$ vs. 0.50. These discrepancies suggest Sakri may be using a different sub-table or different frequency interpolation from Table 7.5-6.
- Sakri's angle calculations (AOD, AOA, ZOD, ZOA) are detailed and geometrically derived, which the other reports do not show as explicitly.

9 Python Code Comparison

Table 10: Python Code Comparison

Code Aspect	**Kagal**	**Ralegaon**	**Pombhurna**	**Sakri**
Code files	2 separate scripts	1 embedded script	1 embedded script	No code listings in report (results shown from code)
Language	Python (NumPy, SciPy)	Python (NumPy)	Python (NumPy)	Python (implied)
Random Seed	Not specified	42	42	Not stated
Link Budget Code	Separate modular	Included in main	Included in main	Not shown
$R_{\max}$ method	Analytic	Binary search	Binary search	Not computed (design-verify)
Channel Sim Code	Separate modular	Single combined	Single combined	Not shown
Code Similarity	Distinct, independent implementation	**Ralegaon and Pombhurna code is identical.** Same variable names, seed(42), comments. Ralegaon code references "Pombhurna."		Independent (no code shown)

Key Findings:

- **Ralegaon and Pombhurna** share identical Python code (same seed, variable names, comments, and even the string "Pombhurna 4G Network Design" in the Ralegaon report).
- **Kagal's** code is independently authored with modular architecture.
- **Sakri does not include any code listings.** It reports "Results from code" sections showing output values (LSPs, cluster delays, cluster powers), but the actual Python code is not reproduced in the report. This means the code cannot be independently verified from the document alone.
- Both Ralegaon and Pombhurna compute $L_{\max} = 133$ dB in their code, yet Ralegaon's report text claims 158 dB — an internal inconsistency.

10 Comprehensive Numerical Comparison

Table 11: All Key Numerical Values — Side-by-Side

Quantity	**Kagal**	**Ralegaon**	**Pombhurna**	**Sakri**
f_c	1800 MHz	1800 MHz	1800 MHz	1800 MHz
P_t	46 dBm	46 dBm	46 dBm	46 dBm
G_{ant} (BS)	18 dBi	18 dBi	17 dBi	Not in budget
Cable Loss	2 dB	2 dB	3 dB	Not in budget
$P_{r,\min}$	-104 dBm	-104 dBm	-94 dBm	-104 dBm
L_{sf}	8 dB	8 dB	8 dB	10 dB
h_{BS} (macro)	25 m	25 m	25 m	25 m
h_{UT}	1.5 m	1.5 m	1.5 m	1.5 m
$L_{\max}$	158 dB	158 dB	133 dB	140 dB
d'_{BP}	452.4 m	452.4 m	(implicit)	900 m (error)
$R_{\max}$ (UMa NLOS)	3680 m	3680 m	843 m	Not computed
Reuse Factor N	1	7	7	7
D/R	1.732	4.583	4.583	4.583
Total sites (final)	9 (7M+2μ)	17 (4N+13U)	10 (8M+1μ+1U)	$\geq$5 (3M+μ+2U)
Macrocell R	800 m	750 m	800 m	800 m
Umbrella R	—	1000 m	1000 m	1200 m
Margin at edge	25.9 dB	27.0 dB	0.89 dB	8 dB / 2 dB
H matrix dim.	2×8	2×4	2×4	N/A
CDL steps	12	6+discussion	6+discussion	6 (code results)
Clusters	20	20	3	19

11 Summary of Differences and Assessment

11.1 Strengths by Report

Table 12: Strengths and Weaknesses Summary

Report	Strengths	Weaknesses
Kagal	• Full 12-step CDL implementation • 2×8 cross-polarised MIMO • Professional LaTeX formatting • Street-level site photography • UMi NLOS model for microcells • Simulation output plots (PDP, APS, $\lvert H \rvert$) • Independent, original code • Clean 7-cell hex ring design	• No GPS coordinates for sites • Does not compute thermal noise floor explicitly
Ralegaon	• GPS coordinates for all 17 sites • Detailed umbrella cell justification • KML-based mapping • SVD and capacity analysis	• Code identical to Pombhurna • Code says $L_{\max} = 133$ dB but report says 158 dB • 13 umbrella cells is unusual • Steps 7–12 are discussion only
Pombhurna	• Most detailed link budget • Explicit hand-calculations for each CDL step • Frequency-dependent DS formula	• Only 3 clusters demonstrated • Very tight margin (0.89 dB) • Steps 7–12 are discussion only
Sakri	• Excellent visual documentation (14 figures) • Iterative design with PL verification at each step • Practical range-based cell radius selection • Correct use of $\max(\mathrm{PL}_{\mathrm{LOS}}, \mathrm{PL}_{\mathrm{NLOS}})$ • Correct $N = 19$ clusters for UMa NLOS • Code output shown for	• Breakpoint distance error (900 m vs. correct 452.4 m) • No antenna gain/cable loss in link budget • No $R_{\max}$ computation • No MIMO channel matrix • No code listings included • Uses UMa (not RMa) for umbrella cells • Single-antenna link model only

11.2 Identical Content Alert

Important Finding: The Python code in the Ralegaon report (by Atharva S Chaudhari) is **identical** to the Python code in the Pombhurna report (by Prasad Patil). All variable names, comments, random seed (`np.random.seed(42)`), system parameters ($G_{\text{BS}} = 17$ dBi, $L_f = 3$ dB, $L_{\max} = 133$ dB), and even the docstring "`Pombhurna 4G Network Design`" are preserved in the Ralegaon report without modification. Step 12 in Ralegaon still references "Pombhurna's semi-urban environment."

11.3 Breakpoint Distance Error in Sakri Report

Technical Error: The Sakri report computes the breakpoint distance as $d_{BP} = \frac{4h_{BS}h_{UT}f_c}{c} = \frac{4\times25\times1.5\times1.8\times10^9}{3\times10^8} = 900$ m. However, 3GPP TR 38.901 specifies that for UMa scenarios, **effective heights** must be used: $h'_{BS} = h_{BS} - h_E$ and $h'_{UT} = h_{UT} - h_E$ where $h_E = 1$ m. This gives $h'_{BS} = 24$ m and $h'_{UT} = 0.5$ m, yielding $d'_{BP} = 452.4$ m. The Sakri report uses the wrong heights, overestimating the breakpoint by a factor of ≈2. While this does not affect the NLOS path loss calculation (which does not depend on d_{BP}), it leads to incorrect LOS/NLOS regime selection near the breakpoint.

11.4 Overall Assessment

1. **Link Budget Approach:** Four different approaches are now visible. Kagal/Ralegaon use an extended budget (158 dB). Sakri uses a simplified budget without antenna gain (140 dB). Pombhurna uses the most conservative approach with penetration/body losses (133 dB). All approaches are defensible, but the omission of antenna gain in Sakri's budget is questionable.
2. **Network Architecture:** Kagal has the most elegant design (7-cell hex ring, no umbrella needed). Sakri demonstrates best practice in iterative design (redesigning umbrella cells when PL constraints fail). Ralegaon's 17-site design is over-engineered. Pombhurna is operationally tight.
3. **Channel Simulation:** Kagal is the only report with a full 12-step CDL implementation producing simulation outputs. Sakri implements Steps 1–6 with code results but uses a single-antenna model. Ralegaon/Pombhurna implement Steps 1–6 with identical code.
4. **Documentation Quality:** Sakri provides the best visual documentation (14 figures covering every design stage). Kagal provides the best technical depth (complete formulas, code, and simulation plots). Ralegaon provides the best geographic documentation (GPS coordinates).
5. **Code Originality:** Kagal's code is independently authored. Sakri does not include code (cannot be verified). Ralegaon and Pombhurna share identical code.
6. **Technical Errors:** Sakri has a breakpoint distance error. Ralegaon has an internal $L_{\max}$ inconsistency (report says 158 dB, code computes 133 dB).